

ABSTRACT

Suicide by hanging from ceiling fans has become a serious issue in hostels, hospitals, and residential buildings. Traditional ceiling fans are designed to support considerable weight for mechanical stability, which unfortunately enables individuals to misuse them for self-harm. Since there is no safety mechanism in existing fan structures, such incidents often occur silently without any preventive action or early warning. The proposed system introduces a modified safety mechanism aimed at detecting abnormal force and preventing critical situations. The design incorporates a mechanical arrangement consisting of a clamp and a roller limit switch placed near the fan rod with a small air gap. Under normal operating conditions, the fan functions without any disturbance, and the system remains inactive. However, when an excessive load is applied to the fan rod, axial displacement occurs, causing the clamp to move downward and trigger the limit switch. This detection mechanism acts as an immediate indicator of abnormal conditions. The signal from the switch is sent to an ESP32 microcontroller, which processes the input and activates the alert system. Once triggered, the system generates an audible alert using a buzzer and simultaneously sends a real-time notification through Telegram using Wi-Fi connectivity. This ensures that both nearby individuals and remote contacts are informed instantly, enabling quick intervention. This combined mechanical and IoT-based approach ensures automatic detection and response without human dependency. The system is cost-effective, easy to install, and can be implemented on existing ceiling fans with minimal modifications, making it highly suitable for improving safety in sensitive environments.

PROBLEM STATEMENT

Suicide by hanging using ceiling fans has become a serious social and public safety concern in hostels, hospitals, and residential environments. Traditional ceiling fans are designed to support significant mechanical load, which unintentionally allows them to be misused for self-harm. In many cases, such incidents occur silently and without immediate detection, making timely intervention difficult. Most existing ceiling fan designs lack any built-in safety mechanism that can detect abnormal force or prevent such incidents. As a result, there is a need for a preventive engineering solution that can detect unusual loading conditions on a ceiling fan and trigger an alert system to notify responsible individuals. The proposed system introduces a safety mechanism that detects abnormal downward force applied to the fan rod. When such a force is detected, the system automatically triggers an alert through an electronic

monitoring unit. This enables immediate notification to guardians, wardens, or security personnel, allowing faster intervention and improving the chances of preventing fatal outcomes. The aim is to enhance safety in sensitive environments such as student hostels, hospitals, and residential buildings through a simple and cost-effective technological solution.

PROPOSED SOLUTION

The proposed solution introduces a safety-oriented system designed to detect abnormal downward force applied to a ceiling fan and immediately trigger alerts to prevent potential suicide attempts. The system combines a mechanical sensing mechanism with an electronic monitoring and communication unit to ensure quick detection and response. The mechanical part of the system consists of a clamp attached to the fan rod and a shelf bracket mounted near the ceiling. A roller-type limit switch is fixed on the bracket in such a way that a small air gap of approximately 1–2 mm is maintained between the clamp and the switch. This air gap ensures that the system does not get falsely triggered due to normal fan vibrations during regular operation. The roller switch is positioned slightly below the clamp to maintain proper alignment and sensitivity. When an abnormal downward force is applied to the fan rod, axial displacement occurs in the rod structure. As a result, the clamp moves downward and presses the roller limit switch. Once the limit switch is activated, it sends an electrical signal to the ESP32 microcontroller, which acts as the central controller of the system. Upon receiving the trigger signal, the ESP32 immediately activates multiple safety responses. First, the system turns off the fan through a relay module to stop its operation. At the same time, a buzzer is activated to generate a local alarm, drawing the attention of nearby individuals. In addition to the local alert, the ESP32 sends an emergency notification message through the Telegram messaging platform using its built-in Wi-Fi connectivity. This message is delivered instantly to predefined contacts such as hostel wardens, security staff, or family members. The system operates automatically and does not interfere with the normal functioning of the ceiling fan under regular conditions.